

inverse transforms and transform of derivatives

all we are doing for the first part is finding the inverse laplace transform

this is the same theorem as 7.1.1 but its inversed you will need this when you are solving the problems:

Theorem 7.2.1 Some Inverse Transforms

- (a) $1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$
- (b) $t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}, n = 1, 2, 3, \dots$
- (c) $e^{at} = \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\}$
- (d) $\sin kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\}$
- (e) $\cos kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + k^2}\right\}$
- (f) $\sinh kt = \mathcal{L}^{-1}\left\{\frac{k}{s^2 - k^2}\right\}$
- (g) $\cosh kt = \mathcal{L}^{-1}\left\{\frac{s}{s^2 - k^2}\right\}$

ex:

$$\mathcal{L}^{-1}\left\{\frac{4}{s} + \frac{6}{s^5} - \frac{1}{s+8}\right\} \rightarrow \mathcal{L}^{-1}\left\{\frac{4}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^5}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+8}\right\}$$

the first can be identified as: $1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$

the second can be identified as: $t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}$

the third can be identified as: $e^{kt} = \mathcal{L}^{-1}\left\{\frac{1}{s-k}\right\}$

$$\mathcal{L}^{-1}\left\{\frac{4}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{6}{s^5}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+8}\right\}$$

$$\downarrow$$
$$4(1) + \mathcal{L}^{-1}\left\{\frac{6}{s^5}\right\} - e^{-8t}$$

• notice the denominator exponent is 5 if we subtract one we get the $n!$ which is 4! which is 24 if we divide 24 by 4 we get 6

$$4(1) + \mathcal{L}^{-1}\left\{\frac{6}{s^5}\right\} - e^{-8t}$$

$$4 + \frac{1}{4}\left(\frac{4!}{s^5}\right) - e^{-8t}$$

$$4 + \frac{1}{4}t^4 - e^{-8t}$$

if we reapply the theorem
we get $\frac{6}{s^5}$

ex#2:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2-2}\right\}$$

• notice how this looks like $e^{ut} = \mathcal{L}^{-1}\left\{\frac{1}{s-u}\right\}$
but you need to manipulate it around
First you would need to get the "s" by itself

$$\mathcal{L}^{-1}\left\{\frac{1}{s} \cdot \frac{1}{s-\frac{2}{s}}\right\} \Rightarrow \boxed{\frac{1}{s} e^{2/5t}}$$

ex#3:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+s-20}\right\} \Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{(s+5)(s-4)}\right\}$$

• you would need to
apply Partial Fractions

$$\left(\frac{A}{(s+5)} + \frac{B}{(s-4)} = \frac{1}{(s+5)(s-4)}\right) \cdot (s+5)(s-4)$$

$$A(s-4) + B(s+5) = 1$$

$$@ s = 4 \Rightarrow A(4-4) + B(4+5) = 1$$

$$B = 1/9$$

$$@ s = -5 \Rightarrow A(-9) + B(-5+5) = 1$$

$$A = -1/9$$

$$\mathcal{L}^{-1} \left\{ \frac{A}{(s+5)} \right\} + \mathcal{L}^{-1} \left\{ \frac{B}{(s-4)} \right\}$$

$$\mathcal{L}^{-1} \left\{ -\frac{1}{9} \frac{1}{s+5} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{9} \frac{1}{s-4} \right\}$$

$$\boxed{-\frac{1}{9} e^{-5t} + \frac{1}{9} e^{4t}}$$

ex # 4

$$\mathcal{L}^{-1} \left\{ \frac{s^2 + 1}{s(s-1)(s+1)(s-2)} \right\}$$

• You need to apply Partial Fractions

$$\left(\frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s+1)} + \frac{d}{(s-2)} \right) \cdot s(s-1)(s+1)(s-2) = \frac{s^2 + 1}{s(s-1)(s+1)(s-2)}$$

$$A(s-1)(s+1)(s-2) + Bs(s+1)(s-2) + Cs(s-1)(s-2) + ds(s+1)(s-1) = s^2 + 1$$

$$\begin{aligned} @ s = 0 & \Rightarrow A(-1)(1)(-2) = 1 \\ & A = 1/2 \end{aligned}$$

$$\begin{aligned} @ s = 1 & \Rightarrow B(1)(2)(-1) = 2 \\ & B = -1 \end{aligned}$$

$$\begin{aligned} @ s = -1 & \Rightarrow C(-1)(-2)(-3) = 2 \\ & C = -1/3 \end{aligned}$$

$$\begin{aligned} @ s = 2 & \Rightarrow d(2)(3)(1) = 5 \\ & d = \frac{5}{6} \end{aligned}$$

$$\mathcal{L}^{-1} \left\{ \frac{A}{s} \right\} + \mathcal{L}^{-1} \left\{ \frac{B}{(s-1)} \right\} + \mathcal{L}^{-1} \left\{ \frac{C}{(s+1)} \right\} + \mathcal{L}^{-1} \left\{ \frac{d}{(s-2)} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{2} \cdot \frac{1}{s} \right\} + \mathcal{L}^{-1} \left\{ -\frac{1}{s-1} \right\} + \mathcal{L}^{-1} \left\{ -\frac{1}{3} \cdot \frac{1}{s+1} \right\} + \mathcal{L}^{-1} \left\{ \frac{5}{6} \cdot \frac{1}{s-2} \right\}$$

$$\boxed{\frac{1}{2} - e^t - \frac{1}{3} e^{-t} + \frac{5}{6} e^{2t}}$$

ex #5:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{2160}{s^7} \right\}$$

$$= t^1 - 3t^6$$

• we notice that the first one/second is $t^n = \frac{n!}{s^{n+1}}$
and if we subtract the exponent from the denominator we get what "n" is but the second one needs a little bit more work we can look at the exponent of the denominator and subtract one to get "n" which is 6! which is equal to 720 if we multiply by 3 we get 2160

ex #6:

$$\mathcal{L}^{-1} \left\{ \frac{5s - 8}{s^2 + 16} \right\} \rightarrow \mathcal{L}^{-1} \left\{ \frac{5s}{s^2 + 16} - \frac{8}{s^2 + 16} \right\}$$

$$\mathcal{L}^{-1} \left\{ 5 \cdot \frac{s}{s^2 + 16} - 2 \cdot \frac{4}{s^2 + 16} \right\}$$

$\Rightarrow$

$$\boxed{5 \cos(4t) - 2 \sin(4t)}$$

$$\bullet \cos(ut) = \frac{s}{s^2 + u^2}$$

$$\bullet \sin(ut) = \frac{u}{s^2 + u^2}$$

ex #7:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 9s} \right\} \Rightarrow \mathcal{L}^{-1} \left\{ \frac{1}{s(s+9)} \right\}$$

• You will need to apply Partial Fractions

$$\left(\frac{A}{s} + \frac{B}{s+9} = \frac{1}{s(s+9)} \right) s(s+9)$$

$$A(s+9) + Bs = 1$$

$$@ s=0 \Rightarrow A(0) + \cancel{B(0)} = 1$$

$$A = 1/9$$

$$@ s=-9 \Rightarrow \cancel{A(0)} + B(-9) = 1$$

$$B = -1/9$$

$$\mathcal{L}^{-1} \left\{ A \frac{1}{s} \right\} + \mathcal{L}^{-1} \left\{ B \frac{1}{s+9} \right\}$$

$$\boxed{\frac{1}{9} - \frac{1}{9} e^{-9t}}$$

ex #8:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^3} \right\}$$

• this is $\mathcal{L}^{-1} \left\{ \frac{n!}{s^{n+1}} \right\}$ but we need to alter it

$$\mathcal{L}^{-1} \left\{ \frac{1}{2} \cdot \frac{1}{s^3} \right\} \rightarrow \boxed{\frac{1}{2} t^2}$$

• to get it to how the given looks we need to add the $1/2$ to it

ex #9:

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^4} \right\}$$
$$= \boxed{\frac{1}{6} t^3}$$

• this is $t^n = \mathcal{L}^{-1} \left\{ \frac{n!}{s^{n+1}} \right\}$ we identified n as 3 and $n! = 3!$ which equal to 6 so to get it to match we add $\frac{1}{6}$

ex #10

$$\mathcal{L}^{-1} \left\{ \frac{1}{s^2} - \frac{48}{s^5} \right\}$$
$$= \boxed{t - 2 t^4}$$

• $t^n = \mathcal{L}^{-1} \left\{ \frac{n!}{s^{n+1}} \right\}$ for both

• for the second one we figured out that the n is 4 and the numerator would be $4!$ which is equal to 24 and multiplying 24 by 2 you get 48

ex #11 :

$$\mathcal{L}^{-1} \left\{ \left(\frac{2}{s} - \frac{1}{s^3} \right)^2 \right\}$$

$$\left(\frac{2}{s} - \frac{1}{s^3} \right) \left(\frac{2}{s} - \frac{1}{s^3} \right)$$

$$= \frac{4}{s^2} - \frac{2}{s^4} - \frac{2}{s^4} + \frac{1}{s^6}$$

$$\mathcal{L}^{-1} \left\{ \left(\frac{4}{s^2} - \frac{2}{s^4} - \frac{2}{s^4} + \frac{1}{s^6} \right) \right\}$$

$$= \boxed{4t - \frac{1}{3} t^3 - \frac{1}{3} t^3 + \frac{1}{120} t^5}$$

ex #12:

$$\int^{-1} \left\{ \frac{(s+1)^3}{s^4} \right\}$$

$$\int^{-1} \left\{ \frac{s^3}{s^4} + \frac{3s^2}{s^4} + \frac{3s}{s^4} + \frac{1}{s^4} \right\}$$

$$\int^{-1} \left\{ \frac{1}{s} + \frac{3}{s^2} + \frac{3}{s^3} + \frac{1}{s^4} \right\}$$

$$= 1 + 3t + \frac{3}{2}t^2 + \frac{1}{6}t^3$$

$$\begin{array}{c} (s+1) \quad (s+1) \quad (s+1) \\ \quad \quad \quad \curvearrowright \\ \quad \quad \quad s^2 + s + s + 1 \\ (s+1) \quad (s^2 + 2s + 1) \end{array}$$

$$= s^3 + 2s^2 + s + s^2 + 2s + 1$$

$$= s^3 + 3s^2 + 3s + 1$$

we are now moving to "how to solve transform of Derivative Problem"

- 1) transform the D.E with the Laplace theorem
- 2) solve the transformed equation for:

$$\mathcal{L}\{y(t)\} = y(s)$$

- 3) inverse transform the solution to find the D.E:

$$\mathcal{L}^{-1}\{y(s)\} = y(t)$$

- 4) apply initial condition if given

transforms of Derivatives

- ① $\mathcal{L}\{f'(t)\} = sF(s) - f(0)$
- ② $\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$
- ③ $\mathcal{L}\{f'''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$

all of the "transform of Derivative" were found by using this theorem

Theorem 7.2.2

$$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{(n-1)}(0)$$

ex #1:

$$y'' - 4y' = 6e^{3t} - 3e^{-t}, \quad y(0)=1, \quad y'(0)=-1$$

$$\mathcal{L}\{y''\} - 4\mathcal{L}\{y'\} = 6\mathcal{L}\{e^{3t}\} - 3\mathcal{L}\{e^{-t}\}$$

$$s^2 y(s) - sy(0) - y'(0) - 4[sy(s) - y(0)] = 6 \cdot \left(\frac{1}{s-3}\right) - 3 \left(\frac{1}{s+1}\right)$$

$$s^2 y(s) - s(1) - (-1) - 4sy(s) + 4(1) = \frac{6}{s-3} - \frac{3}{s+1}$$

$$s^2 y(s) - s + 1 - 4sy(s) + 4 = \frac{6}{s-3} - \frac{3}{s+1}$$

$$s^2 y(s) - 4sy(s) = \frac{6}{s-3} - \frac{3}{s+1} + s - 5$$

$$y(s) [s^2 - 4s] = \frac{6}{s-3} - \frac{3}{s+1} + s - 5$$

$$y(s) = \frac{6}{s(s-3)(s-4)} - \frac{3}{s(s+1)(s-4)} + \frac{s}{s(s-4)} - \frac{5}{s(s-4)}$$

$$y(s) = \frac{6(s+1) - 3(s-3) + s(s-3)(s+1) - 5(s-3)(s+1)}{s(s-3)(s-4)(s+1)}$$

$$\frac{A}{s} + \frac{B}{(s-3)} + \frac{C}{(s-4)} + \frac{d}{(s+1)} = \frac{6(s+1) - 3(s-3) + s(s-3)(s+1) - 5(s-3)(s+1)}{s(s-3)(s-4)(s+1)}$$

$$A(s-3)(s-4)(s+1) + Bs(s-4)(s+1) + Cs(s-3)(s+1) + ds(s-3)(s-4) = 6(s+1) - 3(s-3) + s(s-3)(s+1) - 5(s-3)(s+1)$$

$$@ s=0 \Rightarrow A(-3)(-4)(1) = 30$$

$$A(12) = 30$$

$$A = 5/2$$

$$@ s=3 \Rightarrow B(3)(-1)(4) = 24$$

$$B = -2$$

$$@ s=4 \Rightarrow C(4)(1)(5) = 22$$

$$C = 11/10$$

$$@ s=-1 \Rightarrow d(-1)(-4)(-5) = -12$$

$$d = -3/5$$

$$\int^{-1} \left\{ A \cdot \frac{1}{s} \right\} + \int^{-1} \left\{ B \cdot \frac{1}{s-3} \right\} + \int^{-1} \left\{ C \cdot \frac{1}{s-4} \right\} + \int^{-1} \left\{ d \cdot \frac{1}{s+1} \right\}$$

$$\int^{-1} \left\{ \frac{5}{2} \cdot \frac{1}{s} \right\} + \int^{-1} \left\{ -2 \cdot \frac{1}{s-3} \right\} + \int^{-1} \left\{ \frac{11}{10} \cdot \frac{1}{s-4} \right\} + \int^{-1} \left\{ -\frac{3}{5} \cdot \frac{1}{s+1} \right\}$$

$$= \boxed{\frac{5}{2} - 2e^{3t} + \frac{11}{10}e^{4t} - \frac{3}{5}e^{-t}}$$

ex #3:

$$y' - y = 1, \quad y(0) = 0$$

$$\mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{1\}$$

$$sY(s) - y(0) - Y(s) = \frac{1}{s}$$

$$sY(s) - (0) - Y(s) = \frac{1}{s}$$

$$sY(s) - Y(s) = \frac{1}{s}$$

$$Y(s) [s-1] = \frac{1}{s}$$

$$Y(s) = \frac{1}{s(s-1)}$$

$$\left(\frac{A}{s} + \frac{B}{s-1} = \frac{1}{s(s-1)} \right) s(s-1)$$

$$A(s-1) + Bs = 1$$

$$@ \quad s = 1 \Rightarrow A(1-1) + B(1) = 1$$

$$B = 1$$

$$@ \quad s = 0 \quad A(0-1) + B(0) = 1$$

$$A = -1$$

$$\mathcal{L}^{-1}\left\{A \frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{B \frac{1}{s-1}\right\}$$

$$= \boxed{-1 + e^t}$$

ex #3:

$$2y' + y = 0, \quad y(0) = -3$$

$$2 \int \{y'\} + \int \{y\} = 0$$

$$2(sy(s) - y(0)) + y(s) = 0$$

$$2sy(s) - 2(-3) + y(s) = 0$$

$$2sy(s) + y(s) = -6$$

$$y(s)(2s + 1) = -6$$

$$y(s) = \int \left\{ \frac{-6}{(2s+1)} \right\}$$

$$y(s) = \frac{6}{2} \cdot \frac{1}{s + 1/2}$$

$$y(s) = -3 e^{-1/2 t}$$

ex #4:

$$y' + 6y = e^{4t}, \quad y(0) = 2$$

$$\int \{y'\} + 6 \int \{y\} = \int \{e^{4t}\}$$

$$sy(s) - y(0) + 6y(s) = \frac{1}{s-4}$$

$$sy(s) - 2 + 6y(s) = \frac{1}{s-4}$$

$$sy(s) + 6y(s) = \frac{1}{s-4} + 2$$

$$y(s) (s+6) = \frac{1}{s-4} + 2$$

$$y(s) = \frac{1}{(s-4)(s+6)} + \frac{2}{(s+6)}$$

$$y(s) = \frac{1 + 2(s-4)}{(s-4)(s+6)}$$

$$\frac{A}{(s-4)} + \frac{B}{(s+6)} = \frac{1 + 2(s-4)}{(s-4)(s+6)}$$

$$A(s+6) + B(s-4) = 1 + 2(s-4)$$

$$@ s = 4 \Rightarrow A(4+6) + B(4-4) = 1$$

$$A = 1/10$$

$$@ s = -6 \Rightarrow A(-6+6) + B(-6-4) = 1 + 2(-10)$$

$$B(-10) = -19$$

$$B = 19/10$$

$$\mathcal{L}^{-1} \left\{ A \frac{1}{(s-4)} \right\} + \mathcal{L}^{-1} \left\{ \frac{B}{(s+6)} \right\}$$

$$\mathcal{L}^{-1} \left\{ \frac{1}{10} \frac{1}{(s-4)} \right\} + \mathcal{L}^{-1} \left\{ \frac{19}{10} \frac{1}{(s+6)} \right\}$$

$$= \boxed{\frac{1}{10} e^{4t} + \frac{19}{10} e^{-6t}}$$

ex #5:

$$y' - y = 2 \cos(st) , \quad y(0) = 0$$

$$\cos kt = \frac{s}{s^2 + k^2}$$

$$\mathcal{L}\{y'\} - \mathcal{L}\{y\} = 2\mathcal{L}\{\cos(st)\}$$

$$sy(s) - y(0) - y(s) = 2 \left(\frac{s}{s^2 + 25} \right)$$

$$sy(s) - (0) - y(s) = \frac{2s}{s^2 + 25}$$

$$y(s) (s-1) = \frac{2s}{(s^2+25)}$$

$$y(s) = \frac{2s}{(s^2+25)(s-1)}$$

$$\left(\frac{As+B}{s^2+25} + \frac{C}{(s-1)} = \frac{2s}{(s^2+25)(s-1)} \right) (s^2+25)(s-1)$$

$$(As+B)(s-1) + C(s^2+25) = 2s$$

$$\cancel{As^2} - \cancel{As} - \cancel{Bs} - B + \cancel{Cs^2} + \cancel{25s} = \cancel{2s}$$

$$As^2 + Cs^2 = 0 , \quad -As - Bs + 25s = 2s , \quad B = 0$$

$$A + C = 0$$

$$-A - B = -23$$

$$(23) + C = 0$$

$$-A - (0) = -23$$

$$C = -23$$

$$A = 23$$

$$\int^{-1} \left\{ \frac{As+B}{s^2+2s} \right\} + \int^{-1} \left\{ \frac{C}{(s-1)} \right\}$$

$$\int^{-1} \left\{ 23 \cdot \frac{s}{s^2+2s} \right\} + \int^{-1} \left\{ -23 \cdot \frac{1}{s-1} \right\}$$

$$= \boxed{23 \cos(st) - 23 e^t}$$